

INDEPENDENT PEER-REVIEW PACKET

EC-020

Uniform-r Fixed-q Full-Dimensional Degree-Four Corridor

Cold Review Packet 14 of 20

Dylan M. Roy Mathematical Research Program

Prepared for independent validation - 27 July 2026

This packet is self-contained for mathematical review. Drive access is supplemental, not required to understand the claim and proof.

READ FIRST

You are being asked to review one narrow terminal object, not the wider q0/P0.2/Theorem B research program. The declared terminal label is an internal research-state classification. Your task is to decide whether the exact statement and its proof/evidence are correct within the stated hypotheses and exclusions.

1. Pass A - Independent reconstruction. Read the exact statement, notation, and hypotheses. Attempt your own derivation or falsification before reading the embedded historical review record.
2. Pass B - Source audit. Compare your reconstruction with the complete controlling source text embedded in Part II. Check every dependency, sign, factor, quantifier, scope boundary, and computational claim.
3. Pass C - Hostile review. Run the object-specific focal checks and invent at least one additional mutation or counterexample attempt.
4. Issue one fail-closed verdict: APPROVE; APPROVE WITH CLARIFICATIONS; AMEND REQUIRED; REJECT; or CANNOT VERIFY.

Do not transfer approval to excluded downstream claims. Do not count access to this packet, the author, or another reviewer as independent mathematical evidence.

Part I - Exact Review Object

Packet	14 of 20
Object ID	EC-020
Object class	Uniform deterministic interval theorem
Exact statement	Every point of P01-UNIFORMQ-D4-BOX-001 satisfies the repaired degree-four capture gates for every $0 < r \leq 1/20$, using the fixed-q q/r monotonicity reduction.
Internal terminal label	CLOSED - EXACT UNIFORM-r FIXED-q FULL-DIMENSIONAL DEGREE-FOUR CAPTURE CORRIDOR
Terminal date	2026-07-22
Recommended expertise	Validated numerics / monotonicity / gradient dynamics

1. Significance

This object was separated from larger theorem claims so that its correctness can be assessed without accepting unrelated Gaussian-field, persistence, adjacency, or asymptotic assertions. A positive review supports only the exact statement above and the dependency uses explicitly listed below.

2. Evidence class and prior review

Evidence	Uniform interval-and-monotonicity certificate, nineteen positive gates, five controls, and independent reexecution.
Prior review record	The deterministic mathematics is terminal; later carrier/package reviews do not reopen the lemma.
Binding exclusions	No finite-Q4 mass, exact full-field transfer, numerical Palm constant, P0.1/P0.2, or theorem conclusion.

3. Dependency map and use restrictions

- Inputs: only definitions, hypotheses, and subordinate identities explicitly reproduced in the controlling source text.
- Output: only the exact statement listed in the review-object table.
- Permitted downstream use: as a cited lemma/interface under identical object definitions and hypotheses.
- Forbidden transfer: no approval of a parent theorem, numerical constant, external release, or canonical promotion unless separately reviewed.

- Object-specific exclusions: No finite-Q4 mass, exact full-field transfer, numerical Palm constant, P0.1/P0.2, or theorem conclusion.

4. Object-specific hostile checks

- Verify the fixed-q object and the direction of q/r monotonicity.
- Check all gate minima occur at the claimed endpoint or are enclosed uniformly.
- Attempt to construct an r-dependent counterexample inside the box.
- Attempt at least one additional mutation that changes an orientation, sign, normalization, law, endpoint, or quantifier while leaving the text superficially similar.

5. Proof-obligation ledger

Obligation	Reviewer test	Initial state
O1 Object identity	Definitions, ordering, normalization, and law match the statement.	OPEN FOR REVIEW
O2 Hypotheses	Every hypothesis used is stated and no stronger premise is hidden.	OPEN FOR REVIEW
O3 Derivation	Each algebraic, analytic, topological, or probabilistic step is valid.	OPEN FOR REVIEW
O4 Quantifiers	Uniformity, existence, endpoint, and asymptotic quantifiers are correct.	OPEN FOR REVIEW
O5 Computation	Any executable or interval result is reproducible and non-vacuous.	OPEN FOR REVIEW
O6 Scope firewall	The conclusion is not silently enlarged beyond the exact object.	OPEN FOR REVIEW
O7 Contradictions	No known correction, counterexample, or retained caveat defeats the claim.	OPEN FOR REVIEW

Part II - Complete Controlling Source Text

The following is a paragraph-order-preserving text reproduction of the controlling Drive artifact. Typography is normalized for this packet; mathematical content is not intentionally rewritten. Use the manifest URL for native-file or byte-level checks.

GP-CLS-PROP-024-v1.0 — CONDITIONAL TERMINAL CLOSURE PACKAGE

UNIFORM-r FIXED-q FULL-DIMENSIONAL DEGREE-FOUR CAPTURE CORRIDOR

Artifact ID: GP-CLS-PROP-024-v1.0

Prepared by: GP — OpenAI GPT-5.6 Thinking

Created: 2026-07-22

Class: CLS-PROP — complete additive package pending Dylan M. Roy’s exact terminal approval

Authority: none before human approval

Canonical impact: NONE before approval; afterward only the exact narrow deterministic object below

Queue candidate: EC-020

[[OBJ:GP-CLS-PROP-024-v1.0]] [[QUEUE:EC-020]] [[OBJ:CL-AUD-073-v1.0]] [[STATUS:HUMAN-APPROVAL-READY]]
[[STATUS:NOT-TERMINAL]]

1. EXACT OBJECT PROPOSED FOR CLOSURE

Let P01-UNIFORMQ-D4-BOX-001 be the fixed physical-q, full-dimensional nine-coordinate box declared in GP-DER-114-v1.0. For every separation

$$0 < r \leq 1/20,$$

every point of that box satisfies the repaired degree-four endpoint-typing, selected-branch entry, cone, strip, transit, section, chart-containment, and capture inequalities encoded in the verified GP-DATA-114-v1.1 interval certificate.

The q/r reduction is fixed-q: q lies in [-1.02, -0.98], so $s = q/r \leq -19.6$ throughout the declared r-domain, and every s-dependent lower margin is minimized at $s = -19.6$.

This package closes only that deterministic interval-and-monotonicity certificate.

2. PRIMARY EVIDENCE

- - GP-DER-114-v1.0 — governing derivation and exact box/object definition.
- - GP-AUD-115-v1.0 — same-line adversarial audit and mutation ledger.
- - GP-DATA-114-v1.1 — clean reversible source/receipt capsule, Drive ID 1f__v4_Nn61G0yRzJgDSJymMIT7iMB3hUI6kc6qGtU9U.
- - GP-DATA-114-v1.0 failed staging object, Drive ID 18auUSnBQ5273Scv-hvjx7zc6-TxQC2078EtkID27eRw, preserved as corrupt provenance and excluded from evidence.
- - GP-PRP-131-v1.0 — frozen C1–C11 review surface.

3. INDEPENDENT REVIEW

CL-AUD-073-v1.0 independently reconstructed the clean capsule and executed a separate pure-Python interval implementation with explicit outward rounding.

It verified:

- - source: 9,787 bytes, SHA-256 9d7cb5d6e13579533c5b485775e67c77da83c149c0bf8bbbe7b7bf85aefdb4ef;
- - receipt: 5,047 bytes, SHA-256 8472209ee589f44f1f3c472c7040d183494743a89e21ad9487a1879d192003bc;
- - q/r worst-case reduction $s=-19.6$;
- - monotonicity direction of every s-dependent margin, including both cone faces;
- - all nineteen stored margins strictly positive;
- - minimum margin 0.007485281374238566 up to outward-rounding ulp;
- - all five planted controls fail for their intended reason;
- - the corrupt v1.0 staging capsule was not treated as evidence;
- - fixed-q versus fixed-s scope and all exclusions.

CL-AUD-073 ends with:

APPROVE EC-020 EXACT INTERVAL AND MONOTONICITY SCOPE

The declared line-independence gate is therefore satisfied.

4. AFFECTED DEPENDENCIES UPON TERMINAL APPROVAL

Retire from active attention:

- - repeated reconstruction of GP-DATA-114-v1.1;
- - repeated verification of q/r monotonicity for the declared box;
- - whether all nineteen encoded degree-four interval margins remain positive for every $0 < r \leq 1/20$;
- - whether the five declared controls fail closed;
- - HB-031's narrow EC-020 interval/monotonicity review request;
- - the EC-020 review portion of GP-PRP-131.

Record as available input:

- - the deterministic set inclusion consumed by the finite-Q4 mass object EC-021;
- - fixed-q corridor geometry for future exact-field work, without promoting that future work.

5. UNAFFECTED DEPENDENCIES AND EXCLUSIONS

This closure does not prove or close:

- - Gaussian or Palm mass of the box;
- - finite-Q4 uniform mass except through the separately reviewed EC-021 object;
- - exact full-field transfer;
- - exact endpoint Hessian typing;
- - exact selected-branch adjacency A_r ;
- - a numerical c_A ;
- - P0.1, P0.2, Theorem B, elder-rule pairing, or lifetime laws;
- - correctness of any object outside the encoded interval-gate ledger;
- - theorem promotion, GP-AUTO-034 activation, machine replacement, ballot passage, release, deletion, or destructive action.

The analytic adequacy and broader geometric interpretation of the repaired degree-four sufficient conditions remain governed by their own sources and are not widened by this terminal certificate.

6. PROPOSED TERMINAL LABEL

CLOSED — EXACT UNIFORM-r FIXED-q FULL-DIMENSIONAL DEGREE-FOUR CAPTURE CORRIDOR ON $0 < r \leq 1/20$

7. REQUIRED REGISTER AND LEDGER UPDATES AFTER HUMAN APPROVAL

1. Append EC-020 to the Closure Log with the exact object, source IDs, hashes, CL-AUD-073 review, exclusions, terminal label, and reopening rule.
2. Mark EC-020 **TERMINAL — REMOVED FROM ACTIVE ATTENTION** in the Easy Closure Queue.
3. Mark HB-031's narrow interval/monotonicity review request **SATISFIED / RETIRED**, while preserving mass and exact-field continuation elsewhere.
4. Update Dashboard terminal count and next technical item.
5. Add Evidence Lineage entries for GP-DATA-114-v1.1 and CL-AUD-073.
6. Add Review Independence disclosure identifying CL as the non-GP line and the independent interval implementation.
7. Add a Transition Register entry that changes only the EC-020 deterministic component from reviewed candidate to terminal narrow closure.
8. Preserve the corrupt v1.0 staging object, all source/receipt bytes, all prior queue states, and all correction provenance.

8. HUMAN APPROVAL REQUIREMENT

Dylan M. Roy must explicitly state:

APPROVE EC-020 EXACT INTERVAL AND MONOTONICITY SCOPE

General authorization to continue Drive work, organize files, or pursue research does not substitute for this item-level terminal decision.

9. REOPENING RULE

Reopen only by additive evidence showing one or more of:

- - source or receipt hash mismatch for GP-DATA-114-v1.1;
- - incorrect q/r monotonicity direction;
- - a negative or zero declared interval margin within the frozen box/r-domain;
- - a planted control that does not fail as declared;
- - use of the corrupt v1.0 staging object as evidence;
- - fixed-q/fixed-s coordinate conflation;
- - object or scope mismatch.

Failure of finite-Q4 mass, exact-field transfer, P0.1, P0.2, or a downstream theorem does not reopen this closure unless it contradicts the exact deterministic interval object itself.

10. CURRENT STATUS

PACKAGE: COMPLETE.

PRIMARY EVIDENCE: COMPLETE.

NON-GP REVIEW: PASS — CL-AUD-073-v1.0.

HUMAN APPROVAL: OPEN.

TERMINAL: NO.

END GP-CLS-PROP-024-v1.0

Part III - Terminal-State Reconciliation

1. Closure decision

Exact internal disposition	CLOSED - EXACT UNIFORM-r FIXED-q FULL-DIMENSIONAL DEGREE-FOUR CAPTURE CORRIDOR
Closure date	2026-07-22
Evidence basis	Uniform interval-and-monotonicity certificate, nineteen positive gates, five controls, and independent reexecution.
Independent review already recorded	The deterministic mathematics is terminal; later carrier/package reviews do not reopen the lemma.

The prior closure record is evidence about the research program's internal state, not a substitute for your review. A new material objection should be reported even if the internal label is CLOSED.

2. Correction and reopening rule

Reopen or amend this object if you identify a counterexample, an object-definition mismatch, a sign/factor/quantifier defect, an unreproducible decisive computation, a hidden premise, or a scope claim not supported by the proof. Corrections must be additive and must preserve the historical source and review record.

3. Limits

- No finite-Q4 mass, exact full-field transfer, numerical Palm constant, P0.1/P0.2, or theorem conclusion.
- A correct lemma may still be unusable downstream if another dependency uses a different field, law, coordinate convention, branch mark, normalization, or parameter domain.
- A prior independent review does not reduce your duty to reconstruct the argument; duplicate lineages must not be counted as additional independence.

Part IV - Reproduction Instructions

- Record your name, institution or affiliation, expertise, date, and any prior exposure to this research program.
- Reconstruct the exact statement from Part I without relying on the historical verdict.
- Audit Part II line by line. For computational objects, obtain the raw carrier from the manifest, verify hashes first, inspect source safety, then execute in a clean environment.
- Complete the obligation ledger and hostile checks. Preserve exact counterexamples, code, logs, and environment details.
- Return the verdict form below plus a finding list with severity: critical, material, clarification, editorial, or no issue.

Part V - Reviewer Verdict Form

Reviewer	_____
Affiliation / expertise	_____
Prior exposure	None / limited / substantial - explain: _____
Independent reconstruction	Complete / partial / not possible
Verdict	APPROVE / APPROVE WITH CLARIFICATIONS / AMEND REQUIRED / REJECT / CANNOT VERIFY
Exact scope accepted	_____
Critical or material findings	_____ _____

Reproduction artifacts

Code/log/hash/location: _____

Signature and date

Required narrative: summarize your independent derivation; list every disagreement; state whether each issue affects the theorem statement, only the proof, only reviewability/reproducibility, or only presentation. State explicitly which excluded claims you did not review.

Part VI - Manifest and Provenance

Controlling source: <https://docs.google.com/document/d/1cbLqX5vrhhtrRA4fvESVySoCIc2Vm6XltUqbRTsAFC8/edit>

Closure / terminal record: <https://docs.google.com/document/d/1cbLqX5vrhhtrRA4fvESVySoCIc2Vm6XltUqbRTsAFC8/edit>

Local source snapshot used to construct this packet: GP-CLS-PROP-024-v1.0 — Conditional Closure of the Uniform-r Fixed-q Degree-Four Corridor

Local source snapshot SHA-256 (DOCX export, transport checksum only):
0f411a7f90464310ae609a71986e7ddd3c792983555a5e1dae405dbcfda0dbe9

Authority note: the transport checksum above identifies the locally downloaded Office export used for packet construction. It is not substituted for any marker-delimited frozen-body hash declared inside the source artifact.

Supplemental Drive access may be granted for raw capsules, revision history, comments, exact native exports, and dependency artifacts. The packet is designed so that those materials strengthen reproducibility rather than supply missing mathematical context.